# ALLEN JOE A

VLSI & Semiconductor Engineer | Digital RTL, TCAD & AI for EDA

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## TECHNICAL SKILLS

- Languages: SystemVerilog, Verilog, Python, C, Tcl, Bash
- Digital Design & EDA: AMBA AXI4/APB, DMA Microarchitecture, ModelSim/Questa, OpenROAD, OpenLane
- Device Modeling & TCAD: Synopsys Sentaurus (SDE, SDevice), TFETs, FinFETs, Ferroelectrics (HZO)
- AI for EDA: PyTorch, PyTorch Geometric (GNNs), Computational Lithography (OPC), XAI

## VERIFIED ENGINEERING PROJECTS

### 1. 4-Channel Adaptive Burst DMA Controller with Hardware Performance Monitoring

- Implemented synthesizable SystemVerilog RTL with dynamic burst sizing (INCR4 to INCR16).
- Verified with 6,266 self-checking assertions; achieved 50% burst count reduction on 512-word transfers.
- Architecture: AMBA AXI4 Master / APB Slave, 4 prioritized descriptor channels, non-intrusive counters.

### 2. Gain-Scheduled All-Digital Phase-Locked Loop (GS-ADPLL) & RTL-to-GDS Flow

- Designed dual-mode digital loop filter switching dynamically from fast lock to low-jitter tracking.
- Modeled DCO and phase-frequency detectors in synthesizable Verilog for clock network distribution.

### 3. FE-DGDM-JLTFET Biosensor Device Physics Modelling (Sentaurus TCAD)

- Modeled 10nm junctionless TFET combining HZO negative-capacitance gate with biomolecular cavity.
- Simulated quantum drift-diffusion and band-to-band tunneling; achieved sub-52 mV/dec subthreshold swing.

### 4. GNN-T Edge-Placement Error Prediction for Computational Lithography

- Developed polygon graph neural network surrogate model to accelerate optical proximity correction (OPC).
- Demonstrated 14x inference speedup over optical numerical simulation solvers.

### 5. CNN-Driven Explainable AI for Lithography Hotspot Detection

- Built convolutional neural network layout classifier with Grad-CAM attribution on power grid benchmarks.

## EDUCATION & ACADEMICS

Bachelor of Engineering in Electronics & VLSI Engineering

Core Coursework: VLSI Physical Design, Solid-State Devices, Computer Architecture, Digital RTL, Verification